



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-052 Protected Practice Review

Open only the matching Rxx after a saved Pxx attempt; then close and do changed retry.

Artifact	C3-052
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



Protection rule

This is diagnosis after effort, not first-attempt teaching. Original attempts remain append-only; review completion does not equal mastery.

R01 - Object storage and namespace mental model

OPEN ONLY AFTER P01 IS SAVED.

What strong evidence looks like

No generated part-file name is a business contract; raw evidence is separated from curated state; logical names remain stable.

Baseline repair guidance

What strong evidence shows Strong evidence separates object paths from logical table identity and treats generated file names as internal implementation details. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Design the namespace for a second domain (support tickets) using different physical locations but the same naming principles.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.

R02 - CSV, JSON, Avro and Parquet trade-offs

OPEN ONLY AFTER P02 IS SAVED.

What strong evidence looks like

Decision mentions schema strength, access pattern and consumer/tool compatibility; trade-offs are explicit.

Baseline repair guidance

What strong evidence shows Strong evidence chooses format from access pattern/schema/ecosystem; Parquet is analytical storage, not a universal interchange format. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Choose a format for a mobile-app event archive with nested attributes and a downstream warehouse load.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.



R03 - Columnar storage, statistics and compression

OPEN ONLY AFTER P03 IS SAVED.

What strong evidence looks like

Source and Parquet logical rows match; required fields/types are intentional; projected read returns only requested columns.

Baseline repair guidance

What strong evidence shows Strong evidence contains real Parquet metadata and source-to-output parity.

Projection/row-group evidence is observed, not guessed. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04
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Changed retry

Create a fresh Parquet file from customers_day1.csv and inspect a different subset of columns.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.



R04 - Partitioning strategy and pruning

OPEN ONLY AFTER P04 IS SAVED.

What strong evidence looks like

Partition key is tied to actual filters; expected partition cardinality and small-file risk are documented.

Baseline repair guidance

What strong evidence shows Strong partitioning follows repeated predicates and manageable cardinality. High-cardinality customer/order keys are rejected unless the workload proves otherwise. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Re-evaluate the plan for a workload that queries mostly by service_date rather than region.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.

R05 - Small-files problem and compaction

OPEN ONLY AFTER P05 IS SAVED.

What strong evidence looks like

Compaction trigger has metrics and thresholds; post-rewrite parity checks are explicit.

Baseline repair guidance

What strong evidence shows Strong compaction reasoning starts from measurable file health and ends with row/key/metric parity after rewrite. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words.

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Changed retry

Design a trigger for a table with fewer files but extreme size skew across regions.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.

R06 - Catalogs, metadata and table namespaces

OPEN ONLY AFTER P06 IS SAVED.

What strong evidence looks like

Contract distinguishes logical identity from physical path and includes operational ownership.

Baseline repair guidance

What strong evidence shows A governed table record states purpose, grain, keys, owner, location, partitioning, freshness, quality and retention. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words.

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Changed retry

Write a registration record for support.tickets_current with different ownership and retention.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.



R07 - Lake vs warehouse vs lakehouse + open table-format trade-offs

OPEN ONLY AFTER P07 IS SAVED.

What strong evidence looks like

Decision is conditional, evidence-based and separates table-format semantics from storage file format.

Baseline repair guidance

What strong evidence shows Delta-vs-Iceberg is conditional. Strong reasoning separates Parquet storage from table-format metadata and includes engine/catalog/operations evidence. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04
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Changed retry

Evaluate a second scenario where the primary query engine changes from Spark to Trino.

Pass rule

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R08 - Delta transaction log and ACID guarantees

OPEN ONLY AFTER P08 IS SAVED.

What strong evidence looks like

Committed version observed; row/key parity proven; no "success" claim from file existence alone.

Baseline repair guidance

What strong evidence shows A Delta table is a committed version reconstructed from log actions. Strong evidence includes an observed version and clear data-file vs `_delta_log` distinction. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Create a fresh Delta table for `customers_day1.csv` at a different path.

Pass rule

PASS only when the changed task is reproduced with review closed, validation passes, and you can explain the evidence. A narrow weakness routes to targeted remediation; it never auto-unlocks a fresh full Gate retest.



R09 - Schema enforcement and schema evolution

OPEN ONLY AFTER P09 IS SAVED.

What strong evidence looks like

Schema diff recorded; change approval explicit; downstream/read compatibility considered.

Baseline repair guidance

What strong evidence shows Schema evolution is an owned contract change. Additive fields still need compatibility/downstream checks; broad permissive evolution is not governance. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Classify a fresh change that widens a numeric type and another that renames a required key.

Pass rule

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R10 - MERGE, upsert, update and delete patterns

OPEN ONLY AFTER P10 IS SAVED.

What strong evidence looks like

Target has one row per order_id; expected updates/inserts occurred; rerunning the same delta is materially idempotent.

Baseline repair guidance

What strong evidence shows Before MERGE, source and target key cardinality are proven. After MERGE, updates/inserts and rerun behavior are independently reconciled. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04 PROTECTED REVIEW

Changed retry

Apply a fresh correction to a different order and insert a new order with a different region.

Pass rule

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R11 - Time travel and reproducible historical reads

OPEN ONLY AFTER P11 IS SAVED.

What strong evidence looks like

Historical read is real runtime evidence; version difference is specific; retention implication is stated.

Baseline repair guidance

What strong evidence shows Historical reads are tied to retained table files/history. Strong evidence identifies exact version differences and does not assume unlimited time travel. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04
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Changed retry

Create another table change, then compare the two newest versions without using the earlier comparison text.

Pass rule

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R12 - Medallion/layered architecture and table contracts

OPEN ONLY AFTER P12 IS SAVED.

What strong evidence looks like

Every layer has a distinct purpose/grain; no source evidence is destroyed; consumer dependency is explicit.

Baseline repair guidance

What strong evidence shows Bronze/silver/gold have distinct contracts and consumers. Redundant copies with no semantic/control boundary are weak architecture. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words. FOUR-COURSE SYSTEM | COURSE 3 DATA ENGINEERING | C3-04
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Changed retry

Design layers for support events where gold is a daily SLA dashboard aggregate.

Pass rule

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R13 - Optimization, retention and maintenance strategy

OPEN ONLY AFTER P13 IS SAVED.

What strong evidence looks like

No destructive cleanup occurs without recovery-window check; maintenance has before/after integrity controls.

Baseline repair guidance

What strong evidence shows Compaction and vacuum solve different problems. Destructive retention cleanup is gated by recovery requirements and dry-run/approval evidence. Common weak evidence ■ Only a screenshot/file listing with no row/key/schema/version validation. ■ A copied script with no explanation of grain/contract or failure behavior. ■ A performance/format claim based on popularity rather than observed workload evidence. ■ A destructive retention/cleanup recommendation without recovery-window reasoning. Fresh mastery check Close this review. Change the domain, data, schema, query pattern, partition choice, update, table-format assumption or retention requirement, then solve again from first principles. Explain the result in your own words.

Changed retry

Design a policy for a regulated table requiring 90-day reproducibility and compare it with a low-risk append-only staging table.

Pass rule

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